NBA Game Outcome Predictor

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Introduction

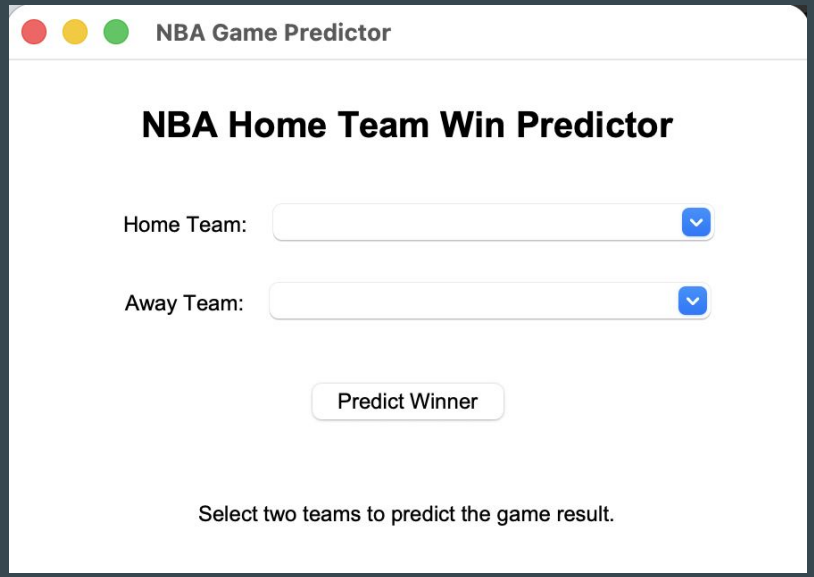
Problem & Background:

- The NBA generates large amounts of game and player data (1,230 games total in a season)
- Outcomes of games are unpredictable due to many influencing factors
- Current predictions often rely on basic stats or subjective analysis
- Opportunities to use data-driven methods for more accurate predictions
- Goal: build a model to predict game outcomes using historical data

Introduction

Hypothesis:

- A model using features like team win percentage, home-court advantage, and recent performance etc, will predict NBA game outcomes with accuracy significantly greater than 50%.
- The accuracy of the model will have an upper bound of around 65%-70% due to the unpredictable nature of sports prediction



Methods

Dataset: NBA API

- Open-source library which downloads statistics from NBA.com
- Provides player stats, team data, and live game updates across multiple seasons
- A python package that is downloadable

Pipeline:

Data Collection → Data Cleaning → Feature Engineering → Modeling → Evaluation

Methods: Tools Used

- Pandas: data manipulation – cleaned, merged, and transformed NBA game and player data
- NumPy: numerical operations
- NBA_api: collected historical, live, and player availability data from the NBA API
- Scikit-learn: trained, compared, and evaluated prediction models
- Tkinter: built the desktop interface for selecting teams and displaying predictions

Methods: Preprocessing

- Data Collection
  - pulled data from 2022-23, 2023-24, and 2024-25 seasons
- Data Cleaning:
  - Ensured only valid NBA teams remained
  - Make sure team names were uniform
  - Removed invalid/missing rows
  - Converted dates → datetime
  - Cleaned raw data by making sure the required columns were there
  - Restructured Data

Methods: Feature Engineering

- Team Strength & Efficiency
  - Net Rating Differential
  - Effective FG% Differential
  - Turnover Rate Differential
  - Star Player Availability (Binary for top 2 players)
- Matchup-Based Features
  - Offensive vs. Defensive Matchup Differentials
- Recent Performance
  - Win % Last 10 Games (Opponent Difficulty Considered)
- Game Context
  - Rest Differential
  - Home/Away
- Player Context
  - Star player availability

Methods: Models

- Logistic Regression
- Random Forest
  - Estimators = 300
  - Max depth = 5
  - Min samples split = 5
  - Min samples leaf = 4
- MLP Classifier
  - Hidden Layer Sizes of 48, 24
  - ReLU
  - Learning rate = 0.001
  - Alpha = 0.01
  - Stopping early = True

Methods

Evaluation methods:

- Accuracy Scores
- Classification Report
  - Precision
  - Recall
  - F1-score
  - Support
- Confusion Matrix

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============================================================
Logistic Regression
============================================================
Training Accuracy: 0.6300090388671287
Test Accuracy: 0.6469879518072289

Confusion Matrix:
[[250 129]
 [164 287]]

Classification Report:
              precision    recall  f1-score   support

           0       0.60      0.66      0.63       379
           1       0.69      0.64      0.66       451

    accuracy                           0.65       830
   macro avg       0.65      0.65      0.65       830
weighted avg       0.65      0.65      0.65       830
```

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============================================================
Random Forest
============================================================
Training Accuracy: 0.6742994877975294
Test Accuracy: 0.6421686746987951

Confusion Matrix:
[[162 217]
 [ 80 371]]

Classification Report:
              precision    recall  f1-score   support

           0       0.67      0.43      0.52       379
           1       0.63      0.82      0.71       451

    accuracy                           0.64       830
   macro avg       0.65      0.63      0.62       830
weighted avg       0.65      0.64      0.63       830
```

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============================================================
MLP Classifier
============================================================
Training Accuracy: 0.6492919554082555
Test Accuracy: 0.6421686746987951

Confusion Matrix:
[[182 197]
 [100 351]]

Classification Report:
              precision    recall  f1-score   support

           0       0.65      0.48      0.55       379
           1       0.64      0.78      0.70       451

    accuracy                           0.64       830
   macro avg       0.64      0.63      0.63       830
weighted avg       0.64      0.64      0.63       830
```

Results: Interpretations, Comparisons

- Logistic Regression performs the best overall on test data
  - Very consistent accuracy → no overfitting
  - Generalizes well
- Random Forest is slightly worse
  - biased toward predicting wins
  - Minor overfitting
- MLP performs nearly as well as Logistic Regression
  - Consistent training and testing accuracy
  - Originally had overfitting problems, but changed parameters and added regularizations

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MODEL COMPARISON SUMMARY
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                Model  Training Accuracy  Test Accuracy
Logistic Regression           0.630009       0.646988
      Random Forest           0.674299       0.642169
     MLP Classifier           0.649292       0.642169
```

Results: Limitations & Future Works

- Does not include current games from the 2025-26 season
- Limited Scope
  - Does not include coaching strategies
- No playoffs data
- Limited player data
  - Star player availability is included, but individual player performance is not
- Assumes past performance = future performance
  - There's a lot that goes into sports beyond what shows up on the stats sheet, so it is very difficult to find truly meaningful features

Results: Real-time Accuracy

For the games on April 10th, 2026 we predicted the following outcomes:
(our predictions are in green, actual winners are underlined)

Suns vs. Lakers

Clippers vs. Trail Blazers

Warriors vs. Kings

Timberwolves vs. Rockets

Grizzlies vs. Jazz

Thunder vs. Nuggets

Magic vs. Bulls

Nets vs. Bucks

Mavericks vs. Spurs

Pelicans vs. Celtics

76ers vs. Pacers

Raptors vs. Knicks

Pistons vs. Hornets

Heat vs. Wizards

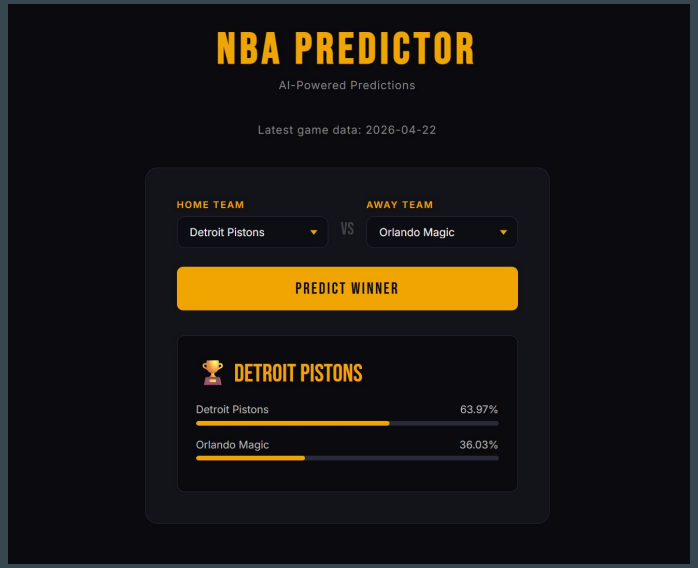
Cavaliers vs. Hawks

12/15 correct predictions = 80% accuracy!

The web hosted version!

Can be accessed from any device at: https://nba-outcome-predictor.vercel.app/

- Runs on the same trained models from our local version
- Has a script for retrieving new/live game data for
- Hosted using Vercel



Acknowledgement

GitHub of nba_api: https://github.com/swar/nba_api/blob/master/README.md

Scikit Learn: https://scikit-learn.org/stable/

Numpy: https://numpy.org/doc/

Pandas: https://pandas.pydata.org/docs/

Tkinter: https://docs.python.org/3/library/tkinter.html

Thank you very much! Questions?